

Nathan (Kaiyan) Zhang

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Education

The University of British Columbia Sep 2023 – May 2028
B.A., Combined Major in Economics and Statistics, Minor in Data Science Cumulative average: 93.5/100
Selected coursework: Statistical Inference (100), Machine Learning and Data Mining (93), Causal Methods (98)
Advanced Econometrics (98), Statistical Learning (97), Time Series and Forecasting (95)

Experience

ESG Co-op (Quantitative Research) Sep 2026 – Present
British Columbia Investment Management Corporation (BCI) / Vancouver, BC
Capital Markets & Credit Investments

- Conduct quantitative analysis of company-level financial and ESG data to support fundamental investment research.
- Assess ESG risks and opportunities and summarize findings for the investment team.

Research Assistant May 2026 – Present
UBC Sauder School of Business | Supervisor: Allen Hu

- Audited RavenPack publication metadata against over 3.2 million ProQuest articles for coverage and data quality.
- Built media collection and validation for a financial-creator directory, preserving raw files and provenance.

Research Assistant Apr 2026 – Present
UBC Sauder School of Business | Supervisors: J. Frank Li and Cheng Chen

- Sole RA on a faculty-led study of age-biased technological change. Linked CPS, PSID, NLSY, O*NET, and PATSTAT to estimate wage and occupational sorting models.
- Harmonized historical O*NET releases and compared task text with about 56M PATSTAT records. Implemented GMM, block bootstrap inference, and an R replication package.

Research Assistant Jan 2025 – May 2026
Vancouver School of Economics at UBC | Supervisors: Jonathan Graves and Laura Nelson

- Built Python/R modules and automated Quarto/GitHub Actions publishing for 1,000+ cumulative student users.
- Designed and taught a computer vision workshop. Coauthored an image-embedding notebook and a Programming Historian tutorial under site-maintainer review.

Selected Research

LLM Evaluation & Memory Efficiency 2026
Manuscript submitted, under review. Co-developed LLM memory evaluation with Jerrold Huang. In a separate 144-item benchmark, PSCC improved exact-match accuracy by about 6.9 percentage points over A-Mem while using roughly one-third fewer API tokens.

Financial NLP & Return Prediction 2026
Manuscript submitted, under review. Built a deduplicated panel of about 31,000 firm events with Jerrold Huang, linking Capital IQ transcripts, IBES, and CRSP. Tested whether management-answer novelty predicts returns beyond earnings surprise using time holdouts, placebo tests, and timing checks.

Causal Analysis of Consumer Spending 2025
Course research project. Applied cross-fitted DoubleML to about 22,000 in-person person-days. Under selection on observables and overlap, estimated a roughly 5% conditional effect of lower-friction payment mixes on daily spending.

Technical Skills

Programming & data: Python, R, SQL, Julia, Databricks, Bloomberg, DuckDB, Polars, Git/GitHub Actions, HPC/Slurm, Jupyter/Quarto
Modeling: causal inference/econometrics, panel/time series, machine learning, NLP, computer vision, scikit-learn, PyTorch, Transformers, fixest, statsmodels
Languages: Mandarin Chinese: native / English: academic working proficiency / Japanese: JLPT N3